

Quantifying and Predicting Flight Delay Propagation Using Statistical Learning Methods

STAT 654: Applied Statistics and Data Analysis

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Abstract

In this study, we investigated how flight delays propagate through consecutive flights of the same aircraft as these planes move between airports over the course of a day. To accomplish this, we analyzed a publicly accessible dataset provided by Kaggle containing information on U.S. domestic flights in 2015. We linked flights together by their unique tail numbers to determine whether delays were transmitted from one flight to the next. Following data cleaning and preparation, we developed features representing possible delay propagation variables, including previous flight arrival delay, turnaround time, airport congestion, and time-of-day factors. The central research question was: to what degree do delays transfer from one flight to another on the same aircraft, and which factors best predict downstream delays? We applied statistical learning techniques from STAT 654, including logistic regression, linear regression, and L1-regularized regression (Lasso), to identify the most influential variables in predicting downstream delays. Our best classification model achieved an area under the curve (AUC) of 0.828 and an accuracy of 86.9%, while our regression models accounted for approximately 32% of the variability in delay minutes. These results confirm that delay propagation is a measurable phenomenon: flights that immediately followed a delayed inbound aircraft experienced a 7-fold increase in the probability of departing late.

1. Introduction

Flight delays are a persistent and costly problem in the U.S. aviation industry. According to the Bureau of Transportation Statistics (BTS), approximately 20% of all commercial flights in the United States experience some form of delay each year, imposing significant financial burdens on airlines, passengers, and the broader economy. Delays arise from numerous sources—weather disruptions, air traffic congestion, mechanical malfunctions, and crew scheduling issues—but one of the least well-understood mechanisms is delay propagation. When an aircraft arrives late at an airport, the next flight it operates is frequently delayed as well, creating a cascading effect that ripples through the network.

Understanding delay propagation is important for multiple stakeholders. For airlines, it directly affects fleet utilization and crew scheduling. For airports, it exacerbates congestion during peak periods and strains gate and runway capacity. For passengers, propagation can result in missed connections, extended wait times, and broader travel disruptions. Despite the scale of this problem, the factors driving delay propagation remain difficult to quantify precisely, in part because they involve complex interactions among aircraft scheduling, airport operations, and external factors such as weather.

This study applied statistical learning methods from STAT 654 to quantify and predict flight delay propagation. We addressed this question from two perspectives. First, we treated it as a classification problem: can we predict whether a flight will be delayed by more than 15 minutes based on its own characteristics and those of its preceding flight? Second, we treated it as a regression problem: can we predict how many minutes a flight will be delayed?

The remainder of this paper is organized as follows. Section 2 describes the dataset and the data preparation process, including the challenges encountered in constructing valid aircraft-level sequence pairs. Section 3 details the feature engineering process. Section 4 presents the exploratory data analysis. Section 5 formalizes the statistical methodology, including the models fitted and the evaluation metrics employed. Section 6 reports the results, and Section 7 discusses the conclusions, limitations, and directions for future work.

2. Data

2.1 Data Source

The dataset used for this project was obtained from Kaggle and contained records of approximately 5.8 million U.S. domestic flights from 2015. The data was originally compiled by the Bureau of Transportation Statistics (BTS). The dataset included a wide range of information for each flight, including airline, flight number, origin and destination airport codes, scheduled and actual departure and arrival times, delay duration, delay cause categories, distance, and the tail number assigned to each aircraft.

The tail number field was particularly important for this analysis because it uniquely identifies each physical aircraft in an airline's fleet. By sorting flights chronologically within each tail number, we constructed sequences of flights operated by the same aircraft and examined how a delay on one flight affected subsequent flights on that plane.

2.2 Data Cleaning

The raw dataset required several preprocessing steps before it could be used for modeling. First, we removed cancelled and diverted flights, as these do not follow standard delay propagation patterns and would introduce noise into the analysis. Second, we addressed missing values by dropping records that lacked departure delay, arrival delay, or tail number fields. Tail number is critical for linking flights into sequences, and departure and arrival delay values are necessary for computing both the response variable and the turnaround time feature. Thirdly, time variables recorded in HHMM format were converted to proper datetime objects to enable accurate computation of turnaround times, including handling overnight flights.

Another major challenge was correctly linking flights at the aircraft level. Each aircraft operates multiple flights per day, and our goal was to connect each flight to the immediately preceding flight on the same aircraft. Initial attempts using a simple sort-and-shift approach resulted in incorrectly matched previous flights, producing impossible negative turnaround times. We implemented additional validation logic to ensure that only valid prior flights were linked, and added code to reset or remove incorrectly sequenced links. These issues underscored the sensitivity of the analysis to proper temporal alignment and data integrity.

Finally, we capped extreme delay values to reduce the influence of outliers. Although some flights experienced delays of many hours, these extreme cases are often driven by unusual circumstances (e.g., mechanical failures or severe weather events) that are not representative of typical propagation dynamics. After all preprocessing steps, the final analysis dataset contained approximately 500,000 flight records with valid propagation linkages.

All preprocessing and feature engineering steps were implemented in a fully reproducible Python pipeline. The workflow was designed to run from raw data ingestion through final dataset construction without manual intervention, and a fixed random seed was used in all model training procedures to ensure reproducibility of results.

3. Feature Engineering

A central component of this project was the construction of features designed to capture delay propagation effects. Rather than relying solely on the raw variables available in the dataset, we engineered several derived features representing the mechanisms through which delays may propagate from one flight to the next.

3.1 Previous Arrival Delay

The most direct measure of propagation is the delay of the immediately preceding flight on the same aircraft. We created a variable called *prev_arrival_delay*, defined as the arrival delay in minutes of the previous flight operated under the same tail number. This variable captures the fundamental premise of the project: if the previous flight arrives late, the current flight is more likely to depart late because the aircraft, crew, and ground operations are not ready for a timely departure.

3.2 Turnaround Time

We created a feature called *turnaround_time*, which measures the elapsed time in minutes between the actual arrival of the previous flight and the actual departure of the current flight. A shorter turnaround time provides less buffer for ground operations—deplaning, cleaning, refueling, and boarding—making it more likely that an inbound delay will propagate to the

outbound flight. This variable was computed using datetime arithmetic after converting the raw HHMM times to proper datetime objects.

3.3 Airport Congestion

We constructed an *airport_congestion* metric defined as the average departure delay across all flights departing from the same origin airport during the same hour of the day. This feature serves as a proxy for the overall operational stress at the airport at the time of departure. Airports experiencing high levels of congestion are more likely to exacerbate individual aircraft delays due to gate conflicts, taxiway queues, and air traffic control restrictions.

3.4 Hour of Day

The *hour* variable extracts the scheduled departure hour (0–23) from the departure time. Because delays tend to accumulate throughout the day as propagation effects compound, flights scheduled later in the day are hypothesized to face greater delay risk. This variable captures time-of-day effects that may interact with congestion patterns.

3.5 Day of Week and Airline

We retained *DAY_OF_WEEK* as a numeric variable (1 = Monday through 7 = Sunday) and *AIRLINE* as a categorical variable with dummy encoding. Different airlines employ different operational strategies, maintain different fleet sizes and hub structures, and build different scheduling buffers into their operations. All of these elements can affect how effectively an airline absorbs or propagates delays. Day of week captures weekly traffic patterns that may influence delay risk.

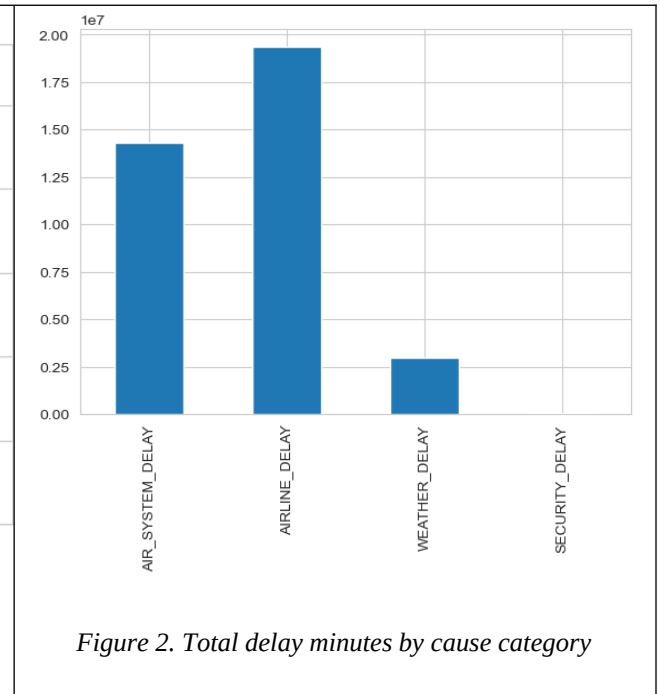
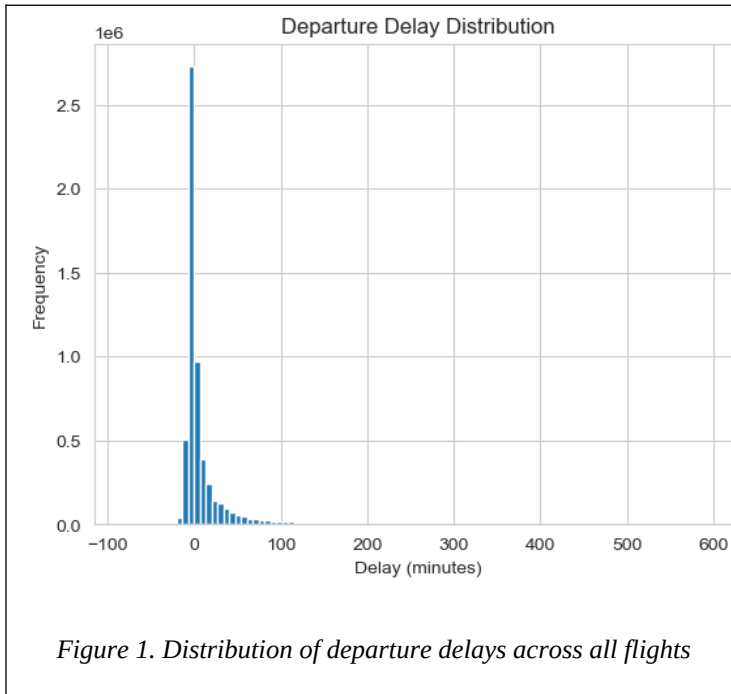
3.6 Response Variables

We defined two response variables corresponding to our two modeling approaches. For classification, we created a binary *delay_indicator* equal to 1 if the departure delay exceeded 15 minutes and 0 otherwise, consistent with the FAA’s standard definition of a delayed flight. For regression, we used the continuous *DEPARTURE_DELAY* variable measured in minutes.

4. Exploratory Data Analysis

4.1 Delay Distribution and Causes

Departure delays are heavily right skewed, with 82.3% of flights classified as not delayed (≤ 15 minutes). As illustrated in Figure 1, This class imbalance (82% non-delayed vs. 18% delayed) is important for model evaluation. Examination of delay causes revealed, as shown in Figure 2, that airline-related delays and air system delays account for the majority of departure delays in the dataset. Weather-related delays and security-related delays are comparatively rare. This finding suggests that most delays are attributable to factors within the airline and air traffic control systems rather than to external disruptions, which supports the premise that propagation—a systemic, operational phenomenon—is a major driver of delay patterns.



4.3 Propagation Evidence

The strongest evidence for delay propagation comes from the conditional probability analysis. When the previous flight on the same aircraft was delayed by more than 15 minutes, the probability of the current flight departing late increased to approximately 70.4%, compared to

only 10.3% when the previous flight was not delayed. This 7-fold increase in delay probability provides compelling evidence that propagation is a dominant factor in determining whether a flight will depart on time.

Figure 3 visualizes this relationship as a scatter plot. The Pearson correlation between *prev_arrival_delay* and *DEPARTURE_DELAY* was 0.469, indicating a moderate positive linear relationship. While this correlation is substantial, the fact that it falls well below 1.0 suggests that the relationship between inbound and outbound delays may be nonlinear—moderate inbound delays may propagate differently than extreme ones—and that additional factors contribute to the variation in flight delays.

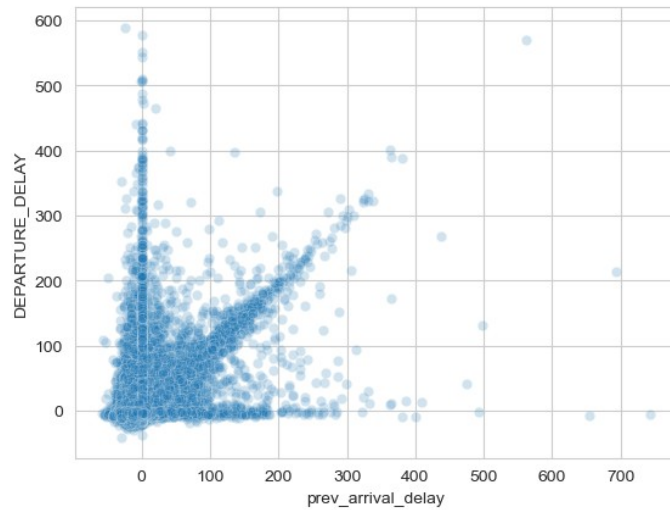


Figure 3. Scatter plot of previous flight arrival delay (x-axis) versus current flight departure delay (y-axis)

4.4 Feature Correlations

A correlation heatmap of all engineered features (Figure 4) confirmed that *prev_arrival_delay* and *airport_congestion* are the strongest correlates of departure delay. The heatmap also revealed that departure delay and arrival delay are highly correlated with each other, as expected, since a late departure typically results in a late arrival. The various delay cause categories (air system, airline, weather, security) exhibited moderate within-group correlations.

turnaround_time showed a weak negative correlation with departure delay, consistent with the hypothesis that longer turnaround windows allow airlines to absorb inbound delays.

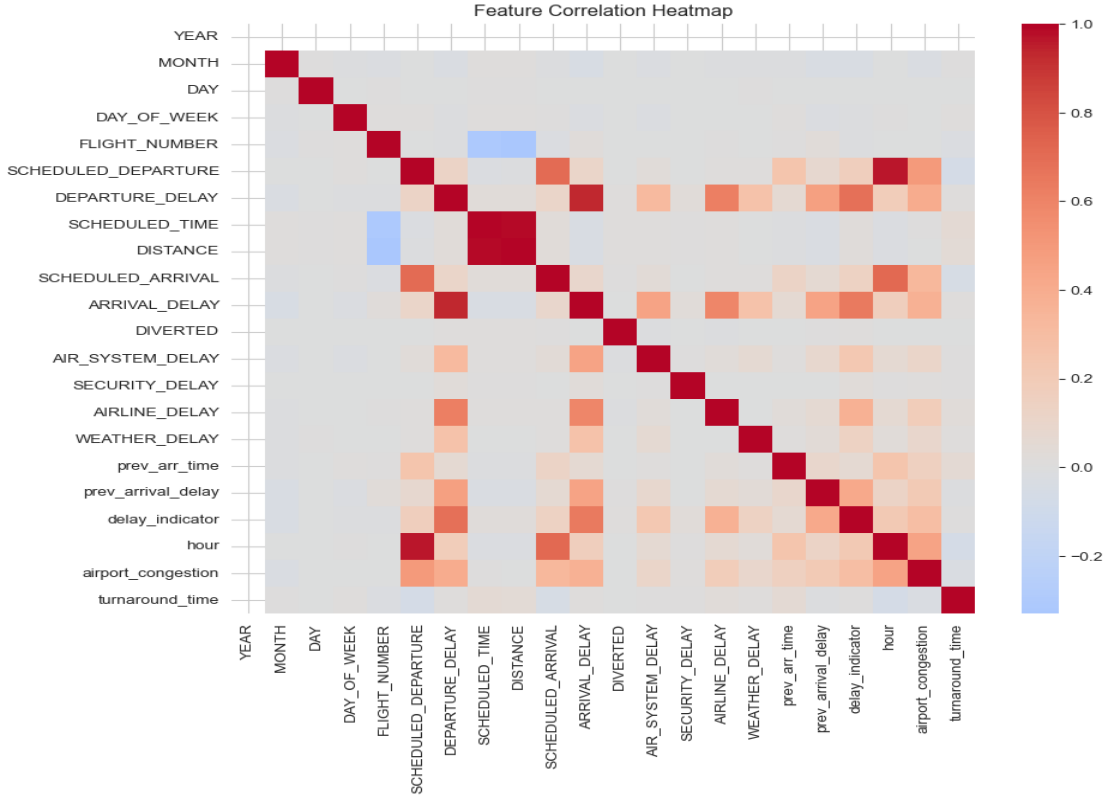


Figure 4. Feature correlation heatmap showing pairwise Pearson correlations among all variables

4.5 Airline-Level Differences

Delay probabilities varied considerably across airlines, as shown in Figure 5. Some carriers exhibited delay rates well above the overall average, while others performed substantially better. We attribute these differences to variation in operational strategies, hub configurations, fleet utilization rates, and scheduling buffer practices. Hawaiian Airlines (HA) and Alaska Airlines (AS) exhibited the lowest delay rates, while United Airlines (UA) and Southwest Airlines (WN) were among the most delay-prone carriers. These airline-level differences supported the inclusion of airline as a categorical predictor in the models. Additionally, Figure 6 illustrates the positive relationship between airport congestion and departure delay, further motivating the inclusion of the airport_congestion feature.

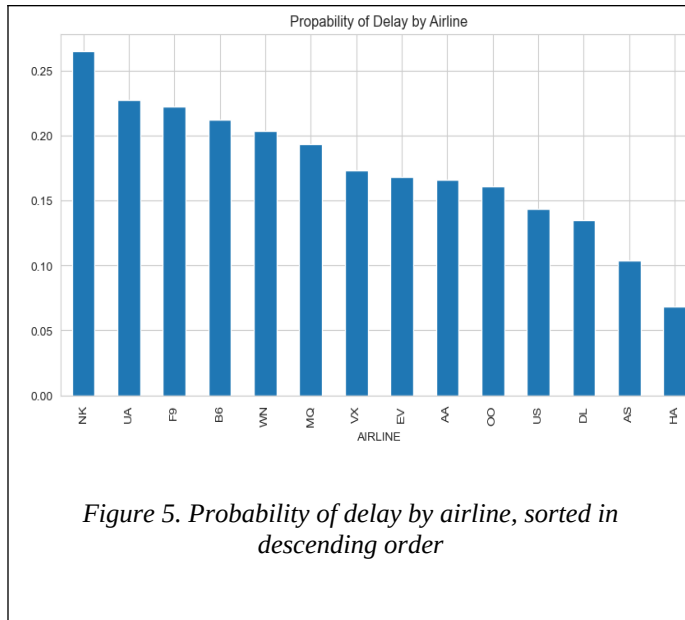


Figure 5. Probability of delay by airline, sorted in descending order

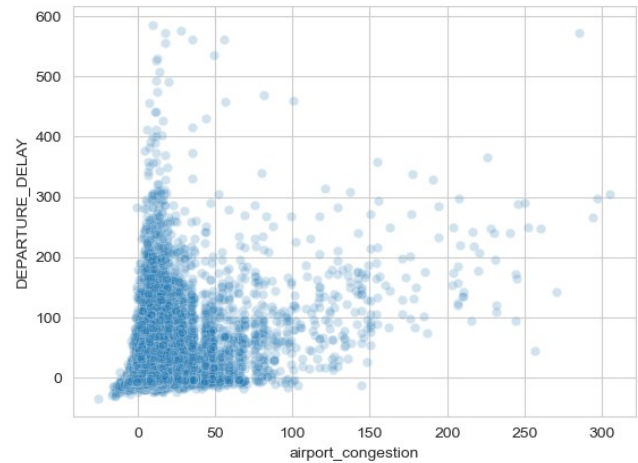


Figure 6. Scatter plot of airport congestion versus departure delay

4.6 Time-of-Day and Day-of-Week Patterns

Delays exhibited distinct time-of-day patterns, as depicted in Figures 7 and 8, with delay risk generally increasing throughout the day as propagation effects accumulated. Flights scheduled in the early morning hours had the lowest delay rates, while afternoon and evening flights were progressively more likely to be delayed. Day-of-week effects were more modest, with slight variation across weekdays. These temporal patterns supported the inclusion of *hour* and *DAY_OF_WEEK* as predictors in the full models.

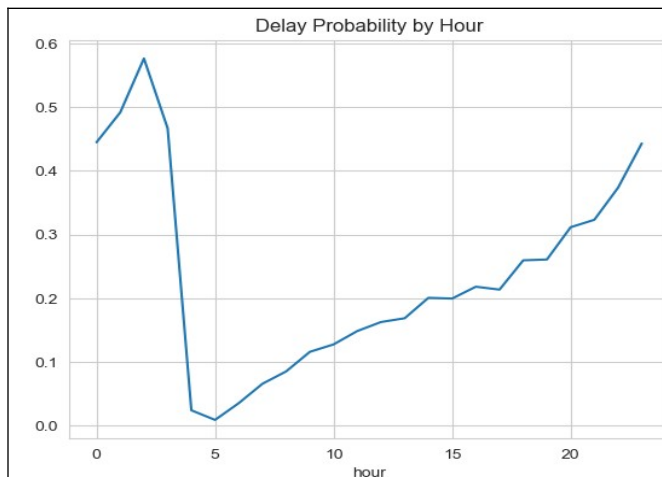


Figure 7. Delay probability by hour of day

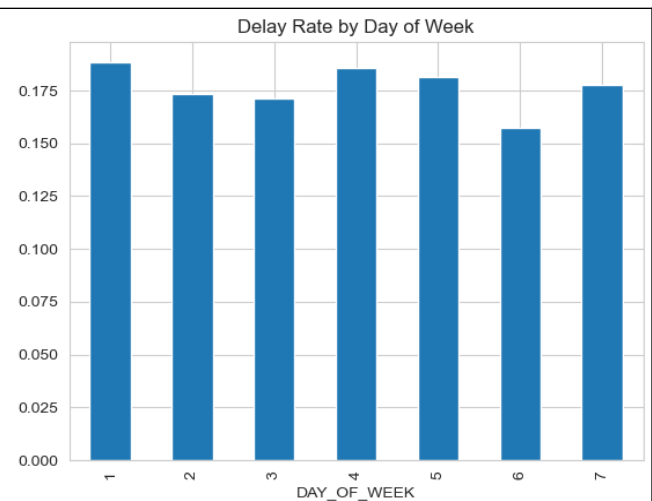


Figure 8. Delay rate by day of week

5. Statistical Methodology

5.1 Research Question Formalization

We examined the research question through two complementary frameworks. In the classification framework, we modeled the probability of a flight being delayed (*delay_indicator* = 1) as a function of the engineered features using logistic regression with a logit link function. In the regression framework, we modeled the expected departure delay in minutes as a linear function of the same predictors using ordinary least squares (OLS). For both approaches, we employed a random 80/20 train-test split to evaluate out-of-sample performance on data not used during model fitting.

5.2 Classification Models

5.2.1 Baseline Logistic Regression

The baseline model included only *prev_arrival_delay* as a predictor. The model takes the form:

$$\log(p / (1 - p)) = \beta^0 + \beta^1 \times \text{prev_arrival_delay}$$

where p is the probability that the flight experiences a departure delay exceeding 15 minutes. This baseline model was developed to quantify the predictive power of the single strongest propagation feature before incorporating additional variables.

5.2.2 Baseline+1 Logistic Regression

The second model added *turnaround_time* as an additional predictor to assess whether the time available for ground operations provides incremental predictive power beyond the inbound delay alone.

5.2.3 Full Logistic Regression

The full model incorporated all engineered features: *prev_arrival_delay*, *turnaround_time*, *airport_congestion*, *hour*, *DAY_OF_WEEK*, and dummy-coded *AIRLINE* indicators (with American Airlines as the reference category). This model was designed to capture the full range of propagation-related and contextual factors simultaneously.

5.2.4 Lasso Logistic Regression

To assess potential overfitting and perform automated variable selection, we applied L1-regularized (Lasso) logistic regression, using cross-validation to select the optimal regularization parameter λ . Lasso penalizes the absolute values of the coefficients, driving unimportant predictors toward zero. If the Lasso model achieves comparable performance to the unregularized full model, this provides evidence that overfitting is not a concern and that all included predictors contribute meaningfully to the prediction.

5.3 Regression Models

5.3.1 OLS Linear Regression

The linear regression model takes the form:

$$DEPARTURE_DELAY = \beta^0 + \beta^1 \times prev_arrival_delay + \beta^2 \times turnaround_time + \dots + \varepsilon$$

where the predictors are identical to those in the full logistic model and ε represents the error term. This model estimates the expected number of delay minutes as a linear combination of the features.

5.3.2 Lasso Linear Regression

We applied Lasso regularization to the linear regression model to validate predictor selection and assess potential overfitting. The Lasso regression uses 3-fold cross-validation to select the optimal λ and evaluates predictive accuracy on the held-out test set.

5.4 Evaluation Metrics

For classification models, we employed the area under the receiver operating characteristic curve (AUC) as the primary evaluation metric because it is robust to class imbalance and measures the model's ability to discriminate between delayed and non-delayed flights across all probability thresholds. We also reported overall accuracy, sensitivity (the proportion of actual delays correctly identified), specificity (the proportion of non-delayed flights correctly classified), and confusion matrices.

For regression models, we used root mean squared error (RMSE) to quantify the average prediction error in minutes and the coefficient of determination (R^2) to measure the proportion of variance in delay minutes explained by the model.

5.5 Model Assumptions and Diagnostics

For the linear regression model, standard assumptions of linearity, independence, homoscedasticity, and normality of residuals were evaluated. Residual plots indicated no severe violations of linearity or heteroscedasticity, and the distribution of residuals was approximately symmetric with mild deviations in the tails, which is expected given the right-skewed nature of delay data.

For the logistic regression models, we assessed the assumption of linearity in the logit for continuous predictors. The relationship between key predictors such as previous arrival delay and the log-odds of delay appeared approximately linear, supporting the use of a logistic regression framework. Independence of observations is assumed given that each flight is treated as a separate unit, though some residual dependence may exist due to unobserved operational factors. Overall, no major violations were identified that would materially affect model conclusions.

6. Results

6.1 Classification Results

6.1.1 Baseline Logistic Model

The baseline model using only *prev_arrival_delay* achieved an AUC of 0.777 and an overall accuracy of 86.3%. The estimated model was:

$$\log(\hat{p} / (1 - \hat{p})) = -1.775 + 0.048 \times \text{prev_arrival_delay}$$

The coefficient of 0.048 indicates that for each additional minute of inbound delay, the log-odds of the current flight being delayed increase by 0.048. This coefficient is highly statistically significant ($p < 0.001$). The confusion matrix revealed that while the model correctly identified 5,047 delayed flights, it missed 12,656 actual delays, yielding a sensitivity of approximately 28%. The high overall accuracy is largely driven by the model's ability to correctly classify the dominant non-delayed class.

6.1.2 Baseline+1 Logistic Model

Adding *turnaround_time* as a second predictor produced a modest improvement, raising the AUC from 0.777 to 0.785. The *turnaround_time* coefficient was small but statistically significant (coefficient = 1.91×10^{-5} , $p < 0.001$), confirming that the time available for ground operations provides some additional predictive power. However, the marginal improvement was modest; turnaround time alone adds limited discrimination beyond what the inbound delay already captures.

6.1.3 Full Logistic Model

The full model, incorporating all 6 predictor categories, achieved the best classification performance with an AUC of 0.828 and an accuracy of 86.9%. All predictors were statistically significant at the 0.05 level. The estimated model included 18 degrees of freedom: 5 continuous predictors plus 13 airline dummy variables.

The confusion matrix for the full model (Figure 9) contained 80,260 true negatives, 6,071 true positives, 1,473 false positives, and 11,502 false negatives. These values correspond to a sensitivity of 34.5% (the model correctly identifies approximately 1 in 3 actual delays) and a specificity of 98.2% (the model is highly accurate at identifying on-time flights). The model is conservative in nature: it rarely predicts a delay when none occurs, but it misses many actual delays due to the class imbalance in the data (82% non-delayed vs. 18% delayed).

Confusion Matrix — Full Logistic Model

Actual	Not Delayed (0)	Delayed (1)
	Not Delayed (0)	Delayed (1)
Not Delayed (0)	80,260	1,473
Delayed (1)	11,502	6,071
	Predicted	

Figure 9. Confusion matrix for the full logistic regression model

Among the predictors, *prev_arrival_delay* remained the most influential feature ($z = 175.5$), followed by *airport_congestion* ($z = 75.7$) and *hour* ($z = 32.1$). Among the airline dummies, United Airlines (UA) had the largest positive coefficient (0.397, $p < 0.001$),

indicating significantly higher delay propensity relative to the baseline carrier (American Airlines). Hawaiian Airlines (HA) had the largest negative coefficient (-0.694 , $p < 0.001$), indicating the lowest delay propensity.

6.1.4 Lasso Logistic Model

The Lasso logistic regression selected an optimal λ of 2.783 via cross-validation. No coefficients were driven to zero, indicating that all predictors in the full model contribute meaningfully to the prediction. The *DAY_OF_WEEK* coefficient was closest to zero at -0.011 , suggesting it is the least influential predictor but is still retained by the model. Figure 10 compares the AUC values across all four classification models, and Figure 11 displays the Lasso coefficient magnitudes. The Lasso model achieved an AUC of 0.830 and an accuracy of 86.9%, virtually identical to the full model, confirming that the unregularized model does not suffer from overfitting. Table 1 summarizes the classification model performance across all 4 specifications.

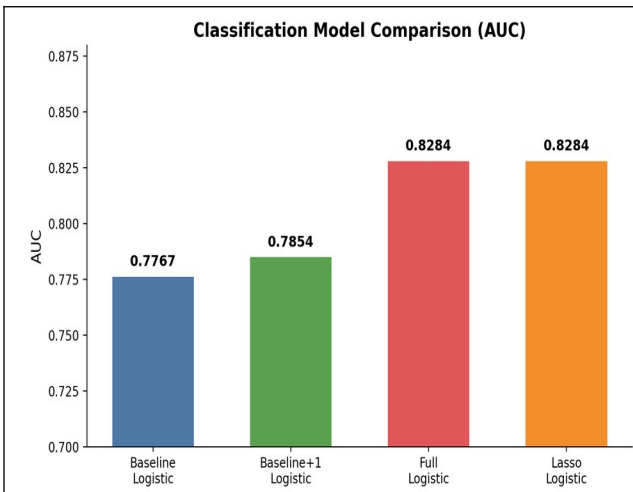


Figure 10. Classification model comparison by AUC

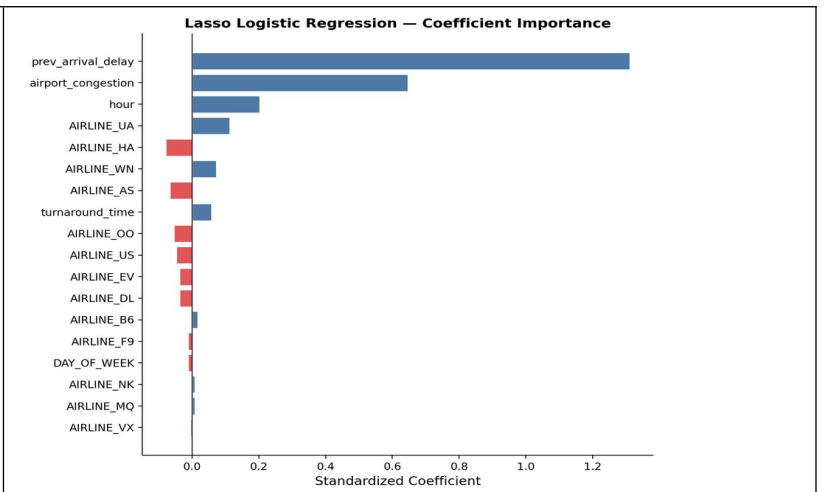


Figure 11. Lasso logistic regression standardized coefficient magnitudes

Table 1: Classification Model Performance Comparison

Model	AUC	Accuracy	Sensitivity	Specificity
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Baseline	0.777	86.3%	28.5%	98.7%
Baseline+1	0.785	86.3%	28.7%	98.7%
Full Model	0.828	86.9%	34.5%	98.2%
Lasso Logistic	0.828	86.9%	34.5%	98.2%

6.2 Regression Results

6.2.1 OLS Linear Regression

The OLS linear regression model achieved an R^2 of 0.319 and an RMSE of 28.15 minutes. The model explains approximately 32% of the variation in departure delay minutes, with the remaining 68% attributable to factors not captured in the dataset. The F-statistic was highly significant (1.036×10^4 , $p < 0.001$), confirming that the predictors jointly possess significant explanatory power.

The *prev_arrival_delay* coefficient of 0.466 indicates that for each additional minute the previous flight was late, the current departure is expected to be approximately 0.47 minutes later, making it the strongest individual predictor. The *airport_congestion* coefficient of 0.779 confirms that higher congestion levels translate directly to longer departure delays.

Among the airline-specific estimates, United Airlines exhibited the largest positive effect (+4.39 minutes relative to American Airlines), while Hawaiian Airlines exhibited the largest negative effect (−2.42 minutes relative to American Airlines), indicating that UA is the most delay-prone carrier and HA is the least delay-prone after controlling for all other predictors.

An interesting finding was that the *hour* coefficient was negative (−0.030, $p = 0.004$) in the regression model, suggesting that later departures are associated with slightly shorter delays in minutes. This contrasts with the positive *hour* coefficient in the classification model (0.041, $p < 0.001$), which indicates that later flights are more likely to exceed the 15-minute delay threshold. This apparent contradiction can be reconciled: while flights departing later in the day may face a somewhat higher probability of being delayed, the magnitude of the delay may decrease as airlines implement recovery strategies over the course of the day.

6.2.2 Lasso Linear Regression

The Lasso linear regression selected an optimal λ of 0.043. Three airline coefficients—JetBlue (B6), Envoy Air (MQ), and Virgin America (VX)—were driven to zero, indicating that these carriers do not differ significantly from the baseline (American Airlines) in terms of delay minutes after controlling for all other predictors. These were the same airlines that were not statistically significant in the OLS model, providing independent validation of their minimal influence on delay.

The Lasso model achieved an RMSE of 28.15 and an R^2 of 0.317, virtually identical to the OLS model ($R^2 = 0.319$). This confirms that the linear regression model is not overfitting and that the selected predictors are appropriate.

Table 2: Regression Model Performance Comparison

Model	RMSE (minutes)	R^2
Linear Regression	28.15	0.319
Lasso Linear	28.15	0.317

Table 3 presents a combined summary of all 6 models fitted in this analysis.

Table 3: Complete Model Performance Summary

Model	Type	AUC	Accuracy	RMSE	R^2
Baseline Logistic	Classification	0.777	86.3%	—	—
Baseline+1 Logistic	Classification	0.785	86.3%	—	—
Full Logistic	Classification	0.828	86.9%	—	—
Lasso Logistic	Classification	0.828	86.9%	—	—
Linear Regression	Regression	—	—	28.15	0.319
Lasso Linear	Regression	—	—	28.15	0.317

7. Discussion

7.1 Key Findings

The results of this analysis provide clear evidence that flight delay propagation is a real, measurable, and statistically significant phenomenon in the U.S. aviation system. The key findings are summarized below.

1. **Propagation is the dominant factor.** Flights following a delayed inbound aircraft have approximately a 70% probability of departing late, compared to only 10% for flights following an on-time arrival. This huge increase represents the primary finding of this research and confirms the centrality of propagation in explaining flight delay patterns.
2. **Previous arrival delay is the strongest predictor.** Across all models—classification and regression, regularized and unregularized—*prev_arrival_delay* consistently emerged as the most influential feature by a wide margin. In the full classification model, its *z*-statistic was 175.5; in the regression model, its *t*-statistic exceeded 298. No other predictor approached this level of importance.
3. **Contextual factors provide meaningful additional predictive power.** Adding airport congestion, time of day, day of week, and airline identity to the models improved the classification AUC from 0.777 to 0.828. While the inbound delay is the dominant signal, the broader operational context—how busy the airport is, what time of day it is, and which airline is operating—also contributes to predicting delays.
4. **The models are not overfitting.** Lasso regularization confirmed that the full models are appropriately specified. In the classification case, no coefficients were driven to zero and performance was identical to the full model. In the regression case, only 3 airline dummies were zeroed out—the same ones that were non-significant in the OLS model—and RMSE was unchanged.

7.2 Limitations

Several important limitations should be acknowledged. The most significant is that the models explain only approximately 32% of the variance in delay minutes ($R^2 = 0.319$). The majority of delay variation is therefore driven by factors not captured in our feature set. Key omitted variables include detailed weather conditions (the dataset contains only aggregate delay cause categories, not specific weather events), mechanical and maintenance issues, crew scheduling constraints, and air traffic control decisions. All these factors are well-documented contributors to flight delays but were not available in the dataset.

The class imbalance in the data (82% non-delayed vs. 18% delayed) limits the classification models' ability to identify delayed flights. Although the models achieve high overall accuracy, their sensitivity is limited to approximately 35%, meaning they miss roughly 2

out of every 3 actual delays. Techniques such as SMOTE oversampling, cost-sensitive learning, or adjusted classification thresholds could potentially improve sensitivity at the expense of some specificity. Finally, this analysis considers only single-hop propagation—how a delay on one flight affects the immediately following flight on the same aircraft. In reality, delays can cascade across multiple sequential flights, and the cumulative effect may be nonlinear. Modeling multi-hop propagation chains would be a natural and valuable extension of this work.

8. Conclusion and Future Work

8.1 Conclusion

This project set out to quantify and predict flight delay propagation using statistical learning methods. Using approximately 500,000 U.S. domestic flight records from 2015, we demonstrated that delay propagation is a measurable and significant phenomenon: the previous aircraft’s arrival delay is the strongest predictor of both whether the next flight will depart late and by how many minutes. Our best classification model achieved an AUC of 0.828 with 86.9% accuracy, while the regression models explained approximately 32% of delay variation with an RMSE of approximately 28 minutes.

The analysis confirms that propagation metrics—particularly the inbound delay from the previous flight—capture the majority of the predictable signal in flight delay data. Contextual factors such as airport congestion, time of day, and airline identity provide additional but more modest improvements. The consistency of results across regularized and unregularized models provides confidence that our findings are robust and not artifacts of overfitting.

8.2 Future Work

Several directions for future work could extend and improve upon this analysis. First, incorporating granular weather data from sources such as METAR or ASOS weather stations could substantially improve predictive power, as weather is a major driver of delays but was only captured at a coarse level in this dataset. Second, exploring nonlinear modeling methods such as random forests, gradient boosting machines, or neural networks could capture complex interactions and nonlinearities that linear models cannot. Third, addressing the class imbalance through techniques like SMOTE oversampling or adjusted classification thresholds could

improve the models' ability to detect actual delays. Fourth, extending the analysis to multi-hop propagation—tracking how delays cascade across 3 or more sequential flights on the same aircraft—would provide a more complete picture of propagation dynamics. Finally, testing these models on more recent data, particularly from the post-COVID era, would help assess the generalizability of the findings.

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